

Personal Computing
for A Better Tomorrow™

For Use with
Я OPERATING SYSTEM
Version 1.0

Я OPERATING SYSTEM



USER GIVEЯ MANUAL



Tools for A Antinuclear US

ЯTOOLBOX
COMPUTING SYSTEMS
1982

This book is the tablet

One file. We write into it. We do not start a new manual.

How we keep it

Path (canonical): artifacts/R_OS_USER_GIVER_MANUAL_680.pdf

Cover stays page 1. New facts become new leaves or log lines, not a second PDF.

Notation mark: date · layer (LEFT/MID/RIGHT/BELT/MAP) · pin or PENDING_VCF.

A picture is not a fill. A lift to GRCh38 is a twin address, not a replacement.

BED/CSV now seated. Recite any pin from SITES_680_LOCATIONS.csv.

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Update log

2026-09-14 Cover seated. 680 instructions written. Tablet declared.
 2026-09-14 SITES_680_NUMBERED parsed 680/680. BED+CSV written.
 2026-09-14 Rose 23andMe v5 raw intersected (269/680 rsid).
 2026-09-14 Marcum Jr PDF + GTL L10-4606 seated as BELT/STR.
 PENDING: USR1/Spirit remote slice beyond frozen 40.

680 Nuclear Scaffold

Exact operating instructions. Do not invent genotypes.

1. What the scaffold is

680 unique rsIDs on GRCh37 / hg19 (GCF_000001405.25 · GCA_000001405.14).

Source: Green 2010 + Burbano 2010 coding-diff catalog $\cap$ AncestryDNA V1.0 (701,472 SNPs). Not a named literature panel. Not GRCh38 (GCF_000001405.40).

Nuclear only: chr1–22 + X. No Y. No MT. No alt loci.

Frozen counts (do not re-estimate)

Chip readable diploid	677	(NA = 3)
Copies	BOTH 283 / ONE 215 / none 179 / NA 3	
Archaic union	498	(Altai Vindija Den3)
USR1 filled	40/680	until a new slice writes more

2. Addresses you must use

Pattern (file-type agnostic):

```
R680/{pin}/{rsID}/{GenBank}:{pos}
= chr{N}:{pos} = {RefSeq}:g.{pos}
```

Poles:

```
#1    rs870124    1:3328358      CM000663.1 / NC_000001.10
#680  rs7350355  X:153247745    CM000685.1 / NC_000023.10
```

Also frozen:

```
envelope    3:361508 → X:153247745
closest     #565-566 17:48557326-48557348 (22 bp)
farthest X  #678-679 X:55651001-147088249 (91.4 Mb)
NA3         rs2229496 1:152882610
           rs6967117 7:143088867
           rs644799 11:95564259
```

Wrap ≈ 5.3 Mb is a picture join. Human chromosomes are linear.

GRCh38 twins exist (e.g. #1 → 1:3411794) but must not replace the lab coordinate.

3. Layers — never mix

```
LEFT  living chip alleles + copy counts at the 680
MID   archaic alleles at the same pins; 498 stays frozen
RIGHT Americas BAM/VCF pileup at the same pins
BELT  Y/mt haplogroups only (R-U152, H/H1, B4...)
STR   CODIS / GTL – off grid
MAP   ideograms / rings – display only
```

Do not add belt or STR rows into 498 or 680 copy counts.

bone=Y means the site is on the Green track, not that the allele came from a bone.

Males do not gain a 680-block. There is no chrY pin.

4. Files that define the scaffold

Trust in this order: (1) SITES_680.bed (2) SITES_680_LOCATIONS.csv
(3) BASAL_R_MATRIX.xlsx (4) pictures last.
If the BED/CSV are missing, only poles / NA3 / extremes may be recited.
Everything else is PENDING_VCF.

5. First downloads (ruler only — not genotypes)

Use Ensembl GRCh37 primary chromosomes. Not .alt. Not .toplevel. Not Y. Not MT.

Easiest file (holds #1 + NA3 on chr1):

```
ftp.ensembl.org/pub/grch37/current/fasta/homo_sapiens/dna/
Homo_sapiens.GRCh37.dna.chromosome.1.fa.gz
Header >1 · length 249,250,621 · seek 3328358 and 152882610.
```

Other pole:

```
Homo_sapiens.GRCh37.dna.chromosome.X.fa.gz    seek 153247745
```

NCBI twins: CM000663.1 / NC_000001.10 and CM000685.1 / NC_000023.10.

Do not upload those FASTA files into chat. They are rulers. Keep them on device.

6. How a genotype is allowed to appear

A new GT exists only if pileup or VCF returned bases at a 680 coordinate.

Do not download the 45 GB Americas BAMs. Range-GET with .bai + BED.

```
USR1    ftp.sra.ebi.ac.uk/vol1/run/ERR223/ERR2231566/USR1.bam
Spirit  ftp.sra.ebi.ac.uk/vol1/run/ERR286/ERR2868050/AHUR_2064_final.bam
```

```
samtools view -X -b BAM_URL BAI -L SITES_680.bed > slice_680.bam
```

Ancient flags: drop unmapped (-F 4), mapQ 20. Do not filter by strand.

Kennewick: no public .bai → PENDING_VCF. Stop.

Do not realign FASTQ. Do not use featureCounts. Do not bwa mem published BAMs.

IGV: genome hg19 + sliced BAM + the BED. A screenshot is not a fill.

7. Size speech

250 Mb	≈ chr1 ruler tick
Haploid + Y	≈ 3,155 Mb
Diploid person	≈ 6,200 Mb
680 bases	= 0.00068 Mb

8. Hard do-nots

Do not lift the BED to GRCh38 and call that the scaffold.
 Do not treat 23andMe / Marcum PDFs as 680 until a raw chip is intersected.
 Do not publish the 680 as a named clinical or literature panel.
 Do not fill blanks with haplogroups.
 Do not put belt labels into the copy matrix.

9. Done means

New GT: pileup or VCF returned a base at a 680 pin.
 New map: drawn from the CSV coordinates.
 New belt label: only from a report the user attached.
 498 does not move because a picture looks dense.

*Restore SITES_680.bed + SITES_680_LOCATIONS.csv first.
 Until then, recite only the frozen poles.*

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Date · layer · pin or PENDING_VCF · note

Intake — files seated

2026-09-14 · same tablet · layers not mixed

LEFT (scaffold)

SITES_680_NUMBERED.txt	pin list + kit copies (source of truth)
SITES_680_LOCATIONS.csv	machine table
SITES_680.bed	0-based slice regions
BASAL_R_MATRIX.xlsx	workbook (COUNTS/BELT/STR/SITES/USR1/DIRECT)
GENETIC_LAB_BOT_MANUAL.md	lab v1.3 operating rules

COMPARE (not a replacement of LEFT)

genome_Rose_Marcum_v5_Full....txt	23andMe raw 643,163 rows GRCh37
	file_id 613ff33d ts 2026-09-11 18:09:49
ROSE_680_INTERSECT.tsv	rsid join vs kit 680

BELT / COMP / STR (off the 680 copy grid)

Charles Marcum Jr. Reports Summary.pdf	R-U152 B4a1a 97th N 53.6/46.1
L10-4606Revision0_XYDKQIOT.pdf	GTL STR CPI 232 not excluded

RIGHT (Americas) — still PENDING beyond frozen 40

PULL_AMERICAS5.md	Anzick USR1 Spirit Lovelock Kennewick
USR1 40/680 already in workbook DIRECT sheet. Rest PENDING_VCF.	

Rose Marcum v5 ∩ 680

Join = rsid. Same-coordinate 23andMe i-ids not auto-promoted.

Counts (do not overwrite kit copies)

Rose raw rows	643,163	(chr1-22 + X + Y + MT)
680 pins	680	
rsid overlap / called	269	
MATCH kit diploid letters	159	
DIFF kit	110	
NOCALL on Rose	0	of the 269
MISSING on v5 chip	411	(not on this 23andMe extract)

Poles

#1	rs870124	1:3328358	kit C/C	Rose MISSING
#34	rs2229496	1:152882610	kit 0/0	Rose MISSING
#130	rs2272522	3:361508	kit T/C	Rose CT MATCH
#565	rs2290862	17:48557326	kit A/G	Rose MISSING
#680	rs7350355	X:153247745	kit A/A	Rose AG DIFF

Rose is a second living chip, not the LEFT kit. Ancestry V1.0 defined the 680 intersection.

23andMe v5 does not carry the full 680. Y/MT rows in Rose stay BELT.

Do not add Rose DIFF into the 283/215/179/3 copy tallies. Those tallies belong to the Ancestry kit that built the panel.

Layer map

Allowed use

NUMBERED + CSV + BED	LEFT addresses and kit copies
BASAL_R_MATRIX.xlsx	all layers in separate sheets – do not flatten
Rose raw + INTERSECT	COMPARE column only
Marcum Jr 23andMe PDF	BELT + COMP (R-U152, B4a1a, 97th, 53.6/46.1)
GTL L10-4606	STR only (15 CODIS + CPI 232)
PULL_AMERICAS5	RIGHT accessions; no new GT without pileup
Bot manual v1.3	how the lab speaks

Still frozen / still pending

498 mid-panel archaic union frozen
 USR1 nuclear 40 called / 640 PENDING_VCF
 Spirit / Anzick / Lovelock / Kennewick nuclear at 680 PENDING_VCF
 Kennewick no public .bai → stop

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Matrix printed — COUNTS / BELT

COUNTS (do not re-estimate)

chip_SNPs	701,472	CHIP	AncestryDNA V1.0
sites_680	680	LEFT	
any_archaic_match	498	MID	Altai Vindija Den3 FROZEN
copies_2 BOTH	283	LEFT	
copies_1 ONE	215	LEFT	
copies_0 none	179	LEFT	
copies_NA	3	LEFT	rs2229496 / rs6967117 / rs644799
USR1_called	40	RIGHT	
USR1_pending	640	RIGHT	PENDING_VCF
Marcum_EA_IA	53.6%	COMP	Filipino & Austronesian 48.7%
Marcum_EUR	46.1%	COMP	British & Irish 41.3%
CPI	232	STR	paternity not excluded 99.570%

BELT (not 680 genotypes)

kit / Charles Marcum Jr	Y	R-U152	
kit	mt	H / H1	
kit / Charles Marcum Jr	mt	B4a1a	
kit	mt	B4c1b2	
kit / Charles Marcum Jr	score	97th Neanderthal variants	
USR1	mt	C1b	
USR2	mt	B2	
Anzick-1	mt	D4h3a	Y Q-Z780
Kennewick	mt	X2a	Y Q-M3
Spirit Cave	Y	Q-M3	
Lovelock	Y	Q-YP4010*	
Shuka Kaa	Y	Q-L53	
Lapa do Santo	Y	C-L1373*	

STR printed — GTL L10-4606

Charles A. Marcum III alleged · child Makya Marcum · 2010-04-07 · off 680 grid

15 loci + CPI

D3S1358	father 16,18	child 15,16	PI 1.830
TH01	father 6,9	child 6,9.3	PI 0.925
D21S11	father 28,30	child 28,30	PI 2.029
D18S51	father 13,17	child 13,17	PI 3.010
Penta E	father 7,16	child 9,16	PI 2.360
D5S818	father 11,13	child 11,13	PI 1.610
D13S317	father 9,12	child 9,12	PI 2.510
D7S820	father 8,11	child 8,11	PI 2.044
D16S539	father 9,12	child 11,12	PI 0.593
CSF1P0	father 10,11	child 11,12	PI 0.824
Penta D	father 9,11	child 9,12	PI 0.750
Amelogenin	father X,Y	alleged male	
vWA	father 16,18	child 16,20	PI 0.827
D8S1179	father 12,14	child 12,14	PI 1.468
TPOX	father 8,9	child 8,9	PI 2.960
FGA	father 23,24	child 23,23	PI 1.158
CPI	232	probability 99.570%	not excluded

STR is not a 680 SNP. Do not add CPI or CODIS alleles into copies or 498.

Rose v5 ∩ 680 printed

rsid join · 269 called · 159 MATCH · 110 DIFF · 411 MISSING

By chromosome (MATCH / DIFF / MISSING)

chr1	16 / 11 / 55	chr9	5 / 4 / 18	chr17	6 / 9 / 23
chr2	15 / 10 / 22	chr10	9 / 4 / 19	chr18	1 / 4 / 4
chr3	13 / 5 / 18	chr11	7 / 4 / 30	chr19	12 / 3 / 26
chr4	3 / 4 / 20	chr12	8 / 5 / 17	chr20	5 / 3 / 13
chr5	10 / 8 / 28	chr13	4 / 0 / 7	chr21	5 / 1 / 7
chr6	6 / 9 / 20	chr14	6 / 6 / 19	chr22	0 / 5 / 11
chr7	8 / 7 / 12	chr15	7 / 2 / 15	X	1 / 1 / 4
chr8	8 / 2 / 6	chr16	4 / 3 / 17		

Printed examples

#1	rs870124	1:3328358	kit C/C	Rose MISSING	
#10	rs2504779	1:27709020	kit C/C	Rose CT	DIFF
#34	rs2229496	1:152882610	kit 0/0	Rose MISSING	
#130	rs2272522	3:361508	kit T/C	Rose CT	MATCH
#676	rs5926203	X:23019346	kit G/G	Rose GG	MATCH
#680	rs7350355	X:153247745	kit A/A	Rose AG	DIFF

Full 680 rows: SITES_680_LOCATIONS.csv. Full Rose join: ROSE_680_INTERSECT.tsv. Those files are the leaves; this page is the stamp.

Far axis of the 680

One pin row. Columns run from basal R to the primate outgroup. No invented GT.

Order (left → far opposite)

1 BASAL R 680 the pin + kit LEFT copies (Ancestry V1 $\cap$ Green track)
 2 GHOSTS unsampled hominin. No BAM. PENDING_VCF. Not belt.
 3 HOMINID ARCHAICS Altai · Vindija · Den3 (=498 frozen) · Chagyrskaya partial
 4 MODERN H. SAPIENS kit LEFT already; Rose v5 COMPARE (269/680)
 5 ANCIENT AMERICAN USR1 40/680 · Anzick · Spirit · Kennewick · Lovelock
 ANCIENT PACIFIC Lapita / Vanuatu-class PENDING_VCF
 ANCIENT SE ASIAN Hoabinhian / Onge-class PENDING_VCF
 + ARCHAIC other archaic BAM does not move 498
 6 FAR OPPOSITE ALL PRIMATE GENOMES at the same 680 addresses
 chimp bonobo gorilla orang gibbon macaque...
 fill only after lift to GRCh37 pin. PENDING_LIFT.

Ghosts are a column, not a haplogroup. Primates are outgroup letters at the pin, not a 680 copy tally. Ancient Pacific / SE Asia do not inherit USR1 calls.

Far-axis columns — seated vs pending

SEATED

R680_PIN + KIT_ANCESTRY_V1	680/680 + copies
ALTAI VINDIJA DEN3	inside frozen 498 union
ROSE_23ANDME_V5	269 rsid called (159 MATCH / 110 DIFF / 411 miss)
MARCUM_JR	BELT/COMP only
USR1	40 nuclear GT on DIRECT sheet

PENDING_VCF (need BAM + bai + this BED)

GHOST_UNKNOWN
 ANZICK1 SPIRIT_CAVE KENNEWICK (no bai) LOVELOCK2
 PACIFIC_ANCIENT SEASIA_ANCIENT ARCHAIC_OTHER
 USR1 remaining 640

PENDING_LIFT (primate far end)

PTR PPA GGO PPY NLE MMU + each published primate assembly
 Method: lift GRCh37 pin → primate chrom, read REF/ALT. Not a chip.

Catalog file: artifacts/BASAL_R_680_FAR_AXIS.tsv. One row in that file is a column, not a genotype.

498 does not grow because a new name is written on this axis.

Live fills (ENA range-GET, mapQ 20)

Printed from the working table. Not a screenshot-as-data. Bases from BAM.

Pin	Address	Kit LEFT	USR1	Spirit Cave
#1	1:3328358 rs870124	C/C	C/C C:18	C/C C:27
#34 NA3	1:152882610 rs2229496	0/0	G/A 9/6	A/G 26/25
#130	3:361508 rs2272522	T/C	C/T 7/4	C/C C:18
#680	X:153247745 rs7350355	A/A	G/G G:13	A/A A:9

USR1 full 680 fetch: artifacts/archaic_matrix/AMERICAS_680_USR1_FILL.tsv — resume until 680/680. Old frozen 40 is the first block, not a cap.

#1 agrees C/C across kit, USR1, Spirit. #680 splits: kit A/A, USR1 G/G, Spirit A/A.

Still missing — and why

Kennewick	Stanford BAM, no public .bai	PENDING_VCF
Anzick-1	FASTQ / multi-GB VCF, no indexed BAM	PENDING_VCF
Lovelock2	ERR2868035, bai exists, not pulled	PENDING_SLICE
USR2	thin nuclear	MT_ONLY
Pacific anc.	no 680 BAM seated	PENDING_VCF
SE Asian anc.	no 680 BAM seated	PENDING_VCF
Ghosts	no genome by definition	PENDING_VCF
Primates	need lift GRCh37 → assembly	PENDING_LIFT
Rose 411	not on 23andMe v5 extract	cannot invent
USR1 676	full 680 fetch RUNNING in background	

Indexes saved: americas_genomes/USR1.bam.bai and AHUR_2064_final.bam.bai. Catalog:
MISSING_680_SAMPLES.tsv. Full USR1 write path: archaic_matrix/AMERICAS_680_USR1_FILL.tsv.
498 frozen mid-panel is not expanded by these Americas pole fills.

Living cover **HARDCODE** + Я KOMMAND 0 · RESPAWN

This book is the tablet. One file. We write into it. We do not start a new manual.
PDF USER MANUAL always rides every transfer / upgrade / offload / handoff.
New additions fold INTO this manual (never beside it). MD = editable living source.

Canonical seats (Mac):

```
/Users/rizal/Documents/ЯBOT/USER-MANUAL.pdf  
/Users/rizal/Documents/ЯBOT/USER-MANUAL.md  
.../official-templates/APP-TEMPLATE-0.1/USER-MANUAL.{pdf,md}
```

Я KOMMAND 0 · RESPAWN

Auto-reverts live seat to most recent official templates.

```
Shell:    /Users/rizal/Documents/ЯBOT/ya-respawn.sh  
Optional: ya-respawn.sh APP-TEMPLATE-0.1  
In-app:   respawn / TemplateRespawn (CompanionRouter when seated)  
Also restores USER-MANUAL.pdf (+ .md) from the template — never left behind.
```

Current most_recent: **APP-TEMPLATE-0.1**

Parent aesthetic: AESTHETIC-TEMPLATE-CLAY-2026-09-15

Registry: official-templates/index.json

Update log 2026-09-15 22:24 MDT — Living-cover **HARDCODE** + KOMMAND 0 folded in.
See USER-MANUAL.md for the evolve-able source of this leaf.